Fully convolutional neural networks

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Image definition

Definition

Here an image is a 2D array of size $p \times q$. Each array element belongs to $\mathbb{R}^d$. The dimension of the value space d, is often called the number of channels of the image.

Examples

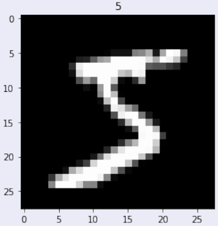
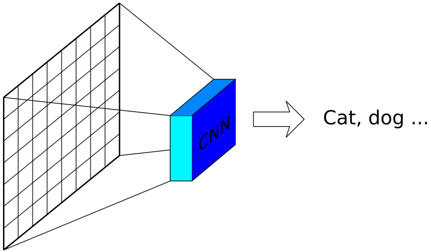


Figure: 28×28 grey level image $(d = 1)$ from the MNIST data set, and 481×321 colour image $(d = 3)$ from the Berkeley segmentation data set.

Learning image transformations

- An image classification task is a function from the set of considered images into a set of labels



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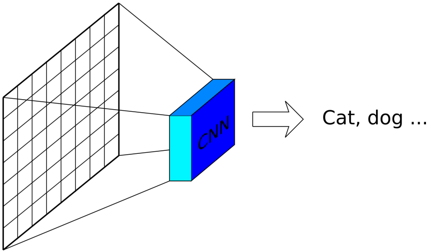


- In many applications, we want to transform an image into another image

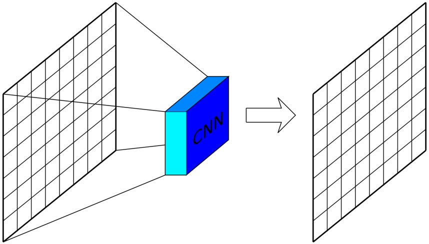
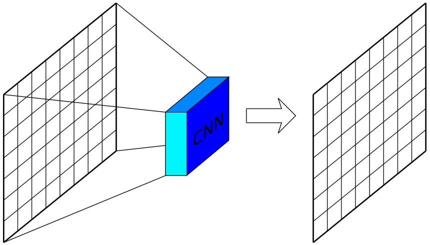


Image-to-image translation

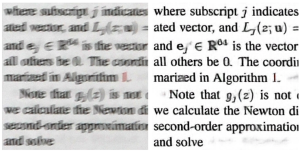
Definition: image-to-image translation

An image-to-image operator is a function that transforms an image into another image. The number of channels and the spatial sizes of input and output images can be different.



Examples

Deblurring network [Hradiš et al., 2015]



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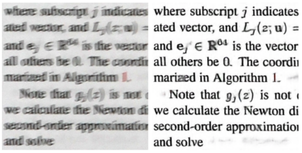
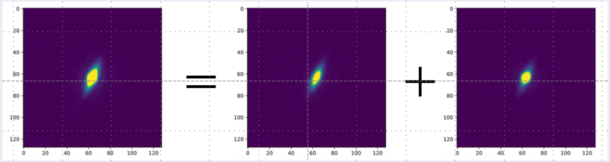


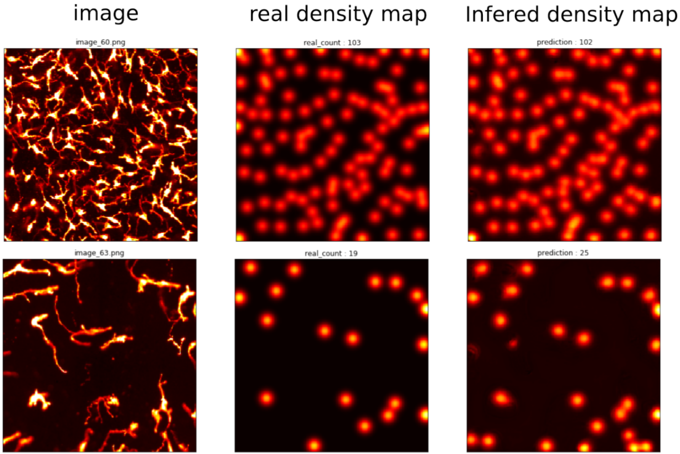
where subscript j indicates
ated vector, and $L_j(z;\mathbf{u})$ =
and $\mathbf{e}_j \in \mathbb{R}^{64}$ is the vector
all others be 0. The coordi
marized in Algorithm 1.
Note that $g_j(z)$ is not
we calculate the Newton di
second-order approximatio
and solve

Bulge / disk decomposition



(Credits: Tuccillo, Huertas-Company, Velasco-Forero, Decencière)

Counting cells



Credits: Tristan Lazard, master thesis.
Images: L'Oréal.

Microscopy cross-modality prediction[Ounkomol et al., 2018]

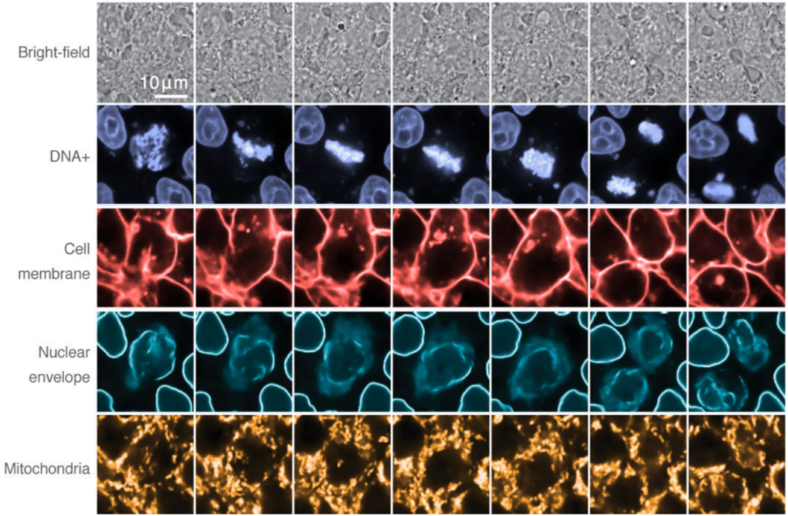
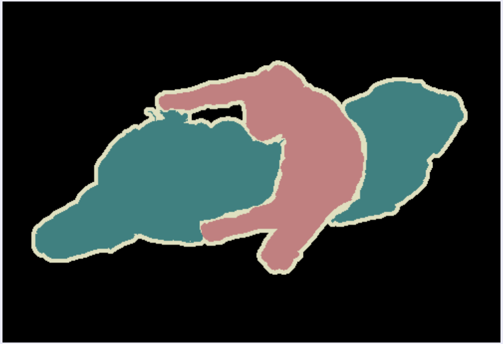


Image segmentation

- Image segmentation is often an important step in an image processing work flow

Example



Credits: Pascal VOC database

Other applications

- Image filtering
- Style modification
- Motion estimation
- etc ...

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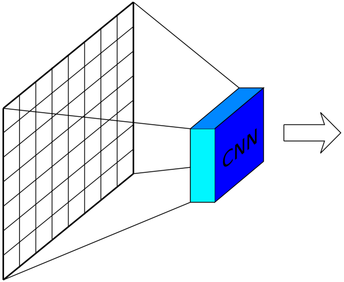
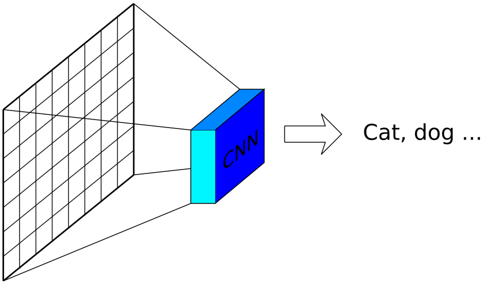
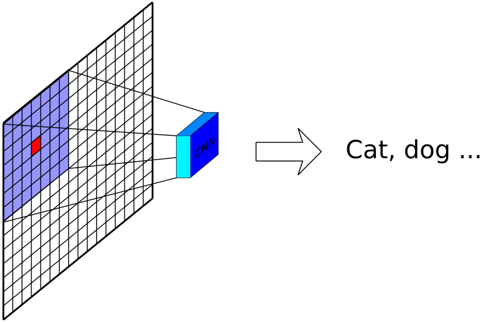
From classification to image-to-image translation



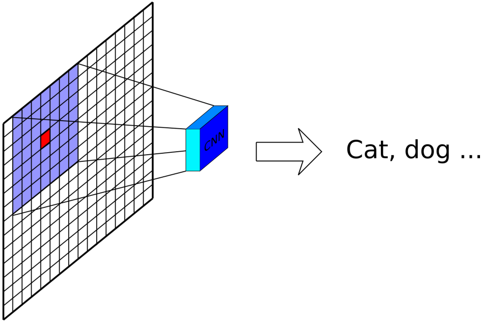
Cat, dog ...

From classification to image-to-image translation





From classification nets to image-to-image nets



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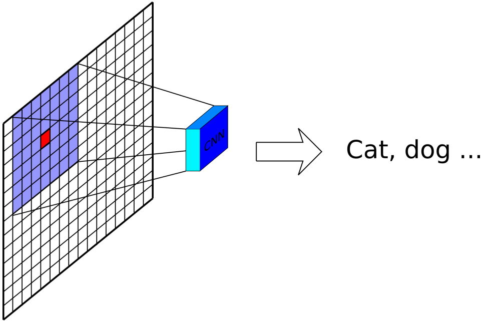

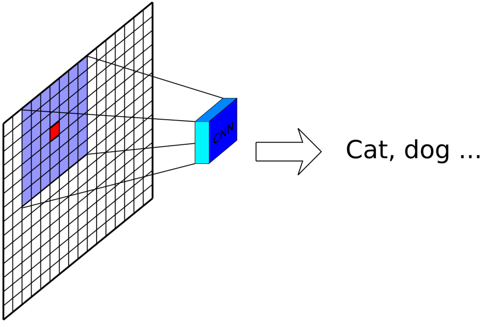

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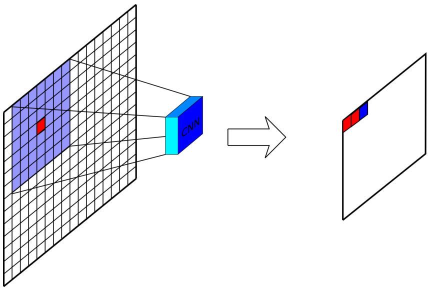

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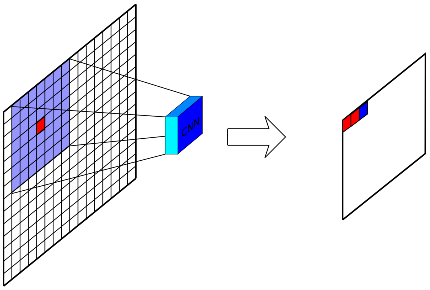
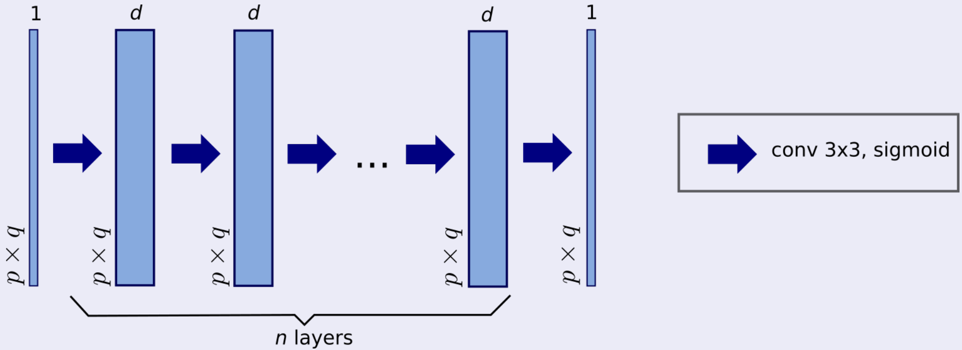


- The neuron membrane segmentation challenge winner [Cireșan et al., 2012] used this strategy. It is inefficient.
- The idea to compute the whole output image with a single pass through the network had already been proposed [Ning et al., 2005, Jain et al., 2007] (but with architectures more complex than those used today).

The simplest image-to-image architecture

Example: plain CNN [Pang et al., 2010]



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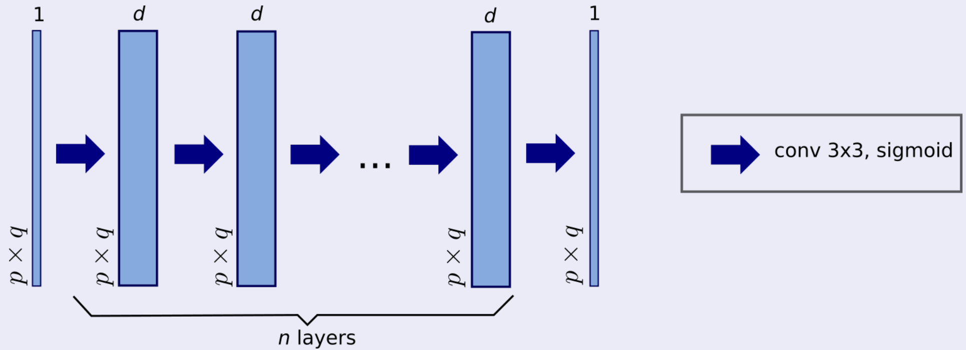

Note that today we would use ReLU activations on all layers except the last.

How many parameters does the first layer contain? The last? The second?

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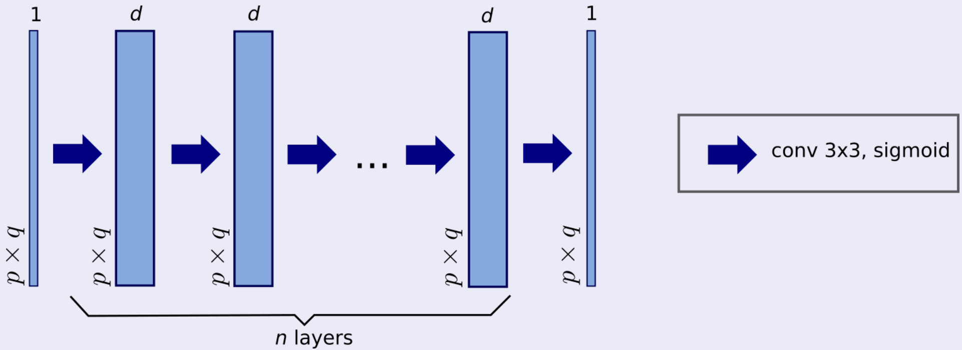
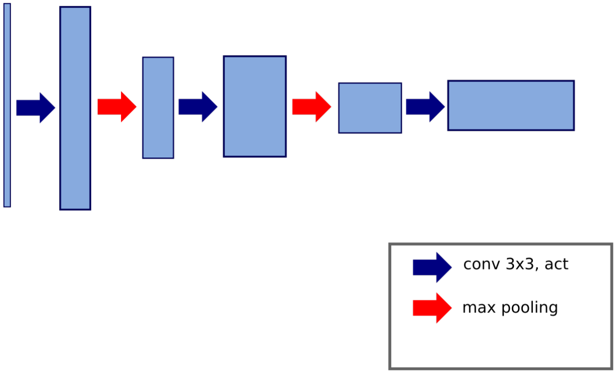


Note that today we would use ReLU activations on all layers except the last.

How many parameters does the first layer contain? The last? The second?

$(9+1) \times d$; $9d+1$; $(9d+1) \times d$

Going back to the original image size



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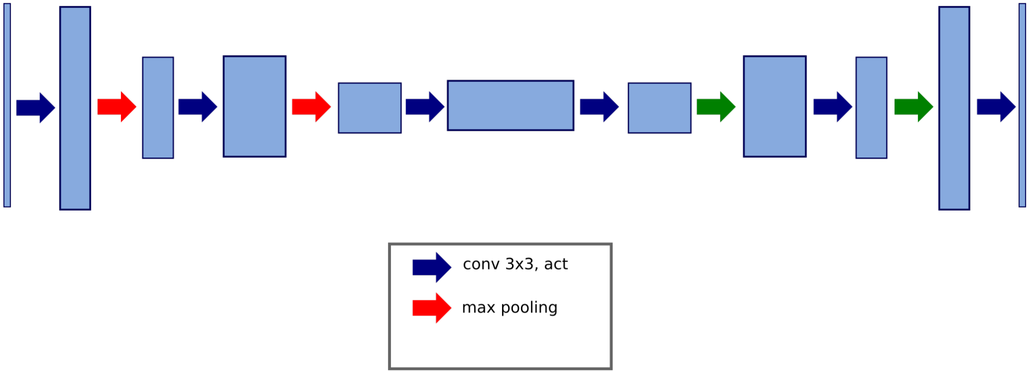

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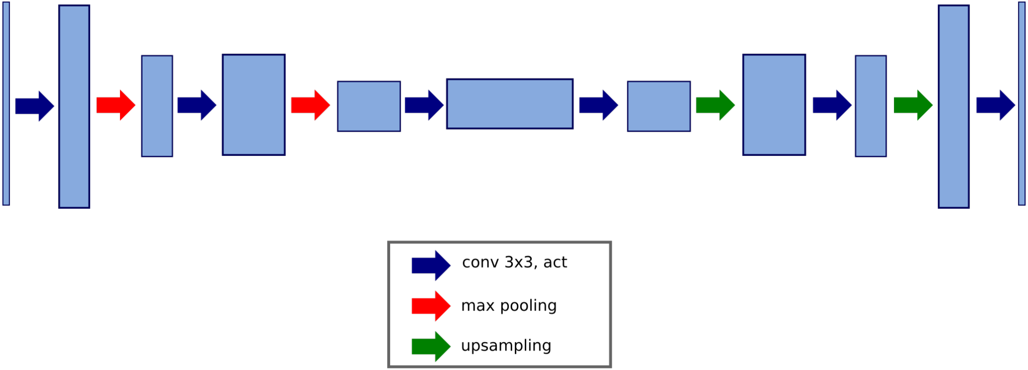
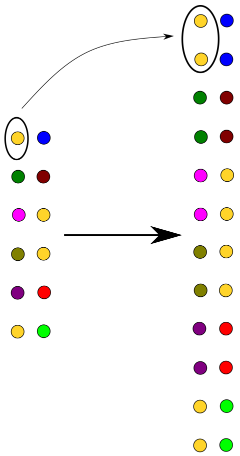


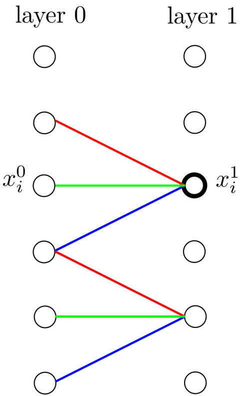
Upsampling techniques

- Replication
- Transposed convolution
- Pooling index memorization

Upsampling through replication

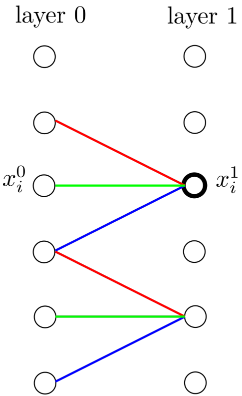


Transposed convolution



Convolutional layer

Transposed convolution



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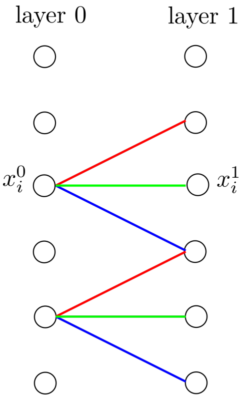

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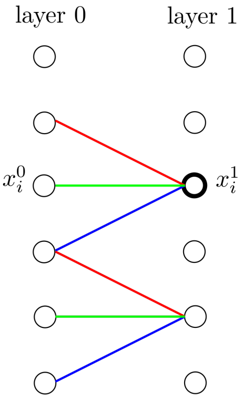

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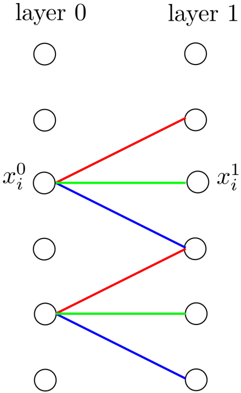
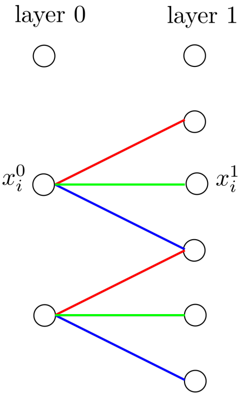


Transposed convolutional layer



Transposed convolutional layer with stride $= 2$

Transposed convolution: example

$$\underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}}_{\textit{Image}} \star \underbrace{\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}}_{\textit{Kernel}} = \begin{bmatrix} 0 & 0 & \\ 0 & 0 & \\ & & \end{bmatrix} + \begin{bmatrix} & 1 & 2 \\ & -1 & 4 \\ & & \end{bmatrix} + \begin{bmatrix} & & \\ 2 & 4 & \\ -2 & 8 & \end{bmatrix} + \begin{bmatrix} & & \\ & 3 & 6 \\ & -6 & 12 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 2 \\ 2 & 6 & 10 \\ -2 & 2 & 12 \end{bmatrix}$$

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Receptive field

Definition: links between neurons

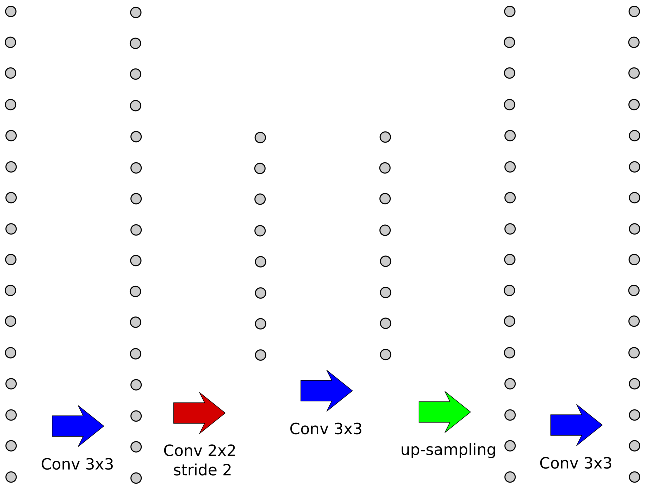
In a NN, we say that neuron a is linked to neuron b if there is an oriented path in the corresponding graph going from a to b.

Definition

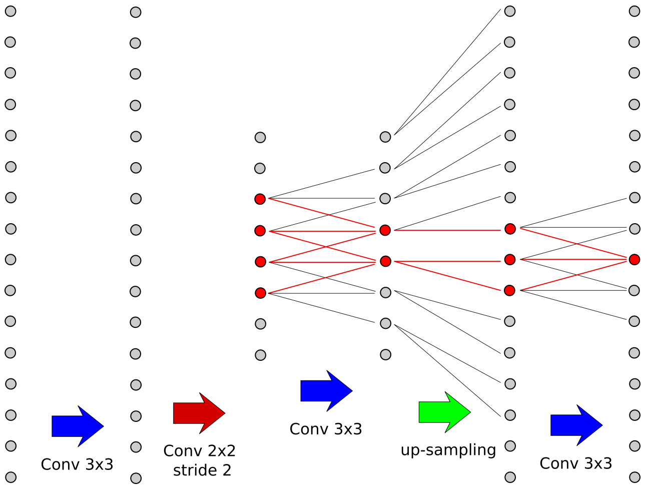
The receptive field of a neuron in a NN is the set of *input neurons* that are linked to that neuron.
The size of the receptive field is an essential property when designing a fully-convolutional NN architecture.

Note that in most cases, receptive fields are square shaped (border effects ignored).

Illustration



Illustration



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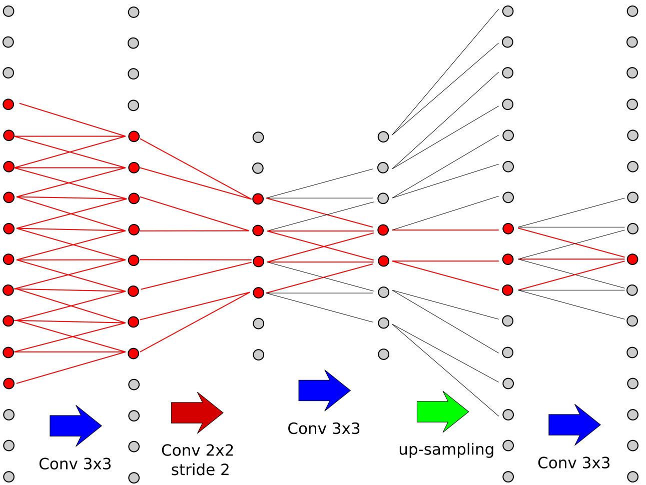
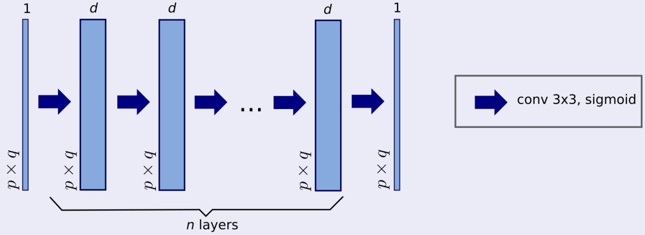


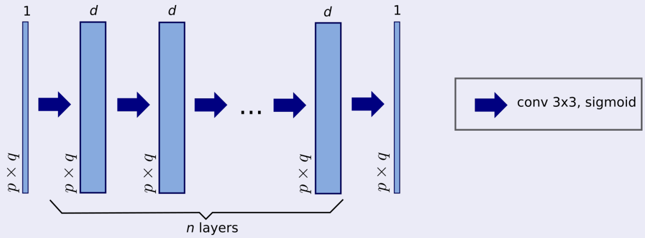
Example



What is the width of the receptive field of the neurons in the last layer?

1. n
2. $1 + 2 \times n$
3. $1 + 2 \times (n + 1)$
4. It depends on the input image

Example



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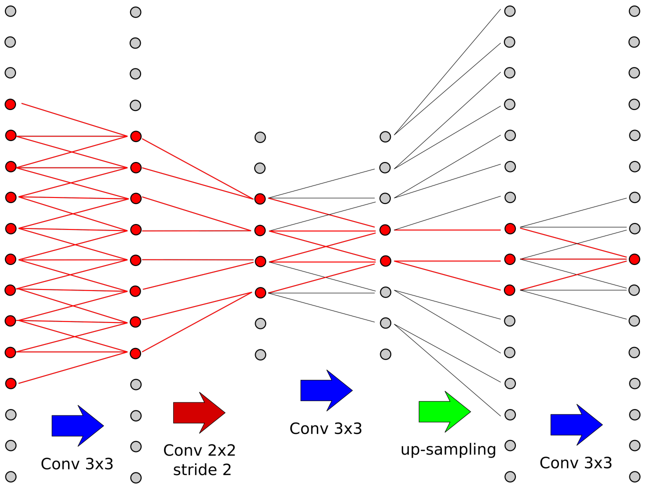
Answer: if the input image is large enough, then its size is $1 + 2 \times (n + 1)$

Effective receptive field: intuition



Criterion

For a given output neuron y, what criterion seems sound to estimate the impact of a pixel $x_{i,j}$ of its receptive field on its value?

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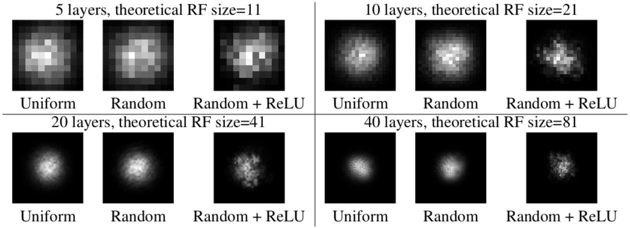
$$\frac{\partial y}{\partial x_{i,j}}$$

Effective receptive field

Shape of the effective receptive field

- Under some simplification hypotheses, the authors of [Luo et al., 2017] have shown that the effective receptive field has a gaussian shape.
- They have experimentally checked that this is (approximately) the case
- Moreover, as the network gets deeper, the size of the effective receptive field gets proportionally smaller



Conclusion on the receptive field

When choosing or designing a NN architecture:

- Establish its minimal required size with respect to the application
- Make it larger than that...

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Equivariance

Definition

A function $f : E \longrightarrow F$ is equivariant with respect to the functions $t_E : E \longrightarrow E$ and $t_F : F \longrightarrow F$ iff $\forall x \in E$:

$$f(t_E(x)) = t_F(f(x))$$

Translation equivariance

- When $E = F$ and t_E and t_F are the same, any, translation, then we have *translation equivariance*.
- Translation equivariance is an often sought property for image processing operators.
- Note that we often abusively say *invariant* to translation instead of *equivariant* to translation.
- Given that in all practical cases images are defined on a bounded set, this property is only true "far enough" from the borders

Quizz

Which operator is NOT translation equivariant?

1. Convolution (stride$=1$)
2. Max-pooling (stride >1)
3. Transposed convolution (stride≥ 1)
4. Upsampling (stride>1)

Translation equivariant operators

Operator	**Translation equivariant**
Convolution (stride$=1$)	
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Translation equivariance: comments

- If padding is used in the network, border effects can be important.
- Translation equivariance is not always welcome!
- Position information can also be used in the network:
  - Through masks or segmentations
  - Through pixel coordinates

Illustration

Credits: OPHDIAT database

Illustration

When aiming to segment the papilla or the macula, translation equivariance is not necessarily welcome.

Credits: OPHDIAT database

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Image size flexibility

- A NN containing fully-connected layers can only process images of a given size.

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- A fully convolutional NN can be applied to images of any size, as long as its dimensions are compatible with the subsampling steps of the network.
- Practical limit: the memory of the system.
- Note that as the input image gets larger, border effects become proportionally less present.

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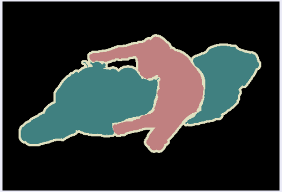
The specific case of image segmentation

Definition: image segmentation

Let I be an image defined on D. A segmentation of I is a partition of D. In practice the regions of the segmentation should correspond to the objects in I, which is application dependant.

- A partition is often represented as a labelled image
- In order to make the segments symmetric, each one is represented by a different channel

Image segmentation example



Credits: Pascal VOC database

Some vocabulary on segmentation

- **Object detection / localization**: bounding box around the object(s).
- **Binary segmentation**: segmentation in 2 classes, background and object.
- **Semantic segmentation**: a label is given to each pixel, according to the object it belongs to.
- **Instance segmentation**: identify each separate object, even if they belong to the same class.

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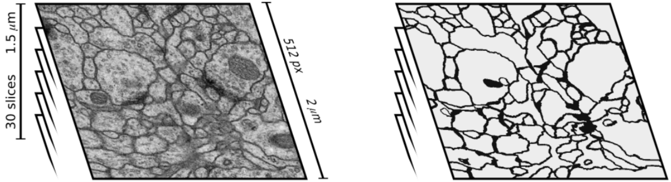
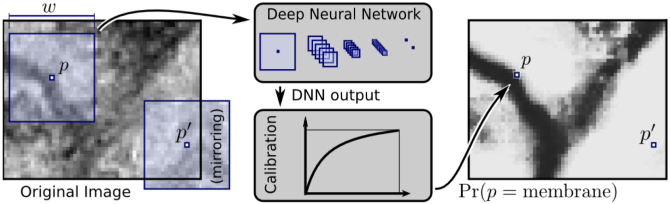
Neuron membrane segmentation challenge (ISBI 2012)

- Train: single stack of size $30 \times 512 \times 512$.
- Test: a second stack of same size.



Neuron membrane segmentation challenge winner [Cireșan et al., 2012]



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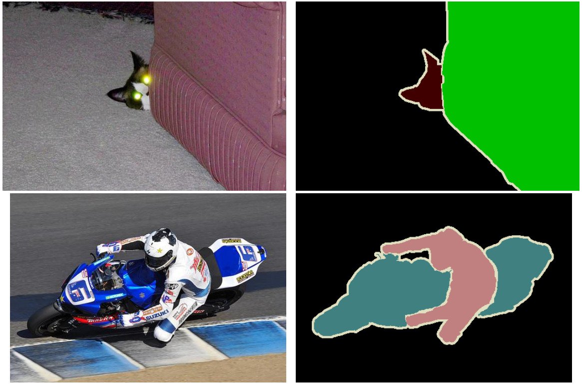
Pascal visual object classes segmentation challenge 2012 [Everingham et al., 2014]

- 1464 training and 1449 validation images
- automatic online test, with unknown images
- 20 image categories (cat, sofa, motorbike, person, etc.)



Convolutional nets for semantic image segmentation

Three papers in 2015:

- Fully convolutional networks for semantic segmentation [Long et al., 2015]
- U-Net: convolutional networks for biomedical image segmentation [Ronneberger et al., 2015]
- SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation [Badrinarayanan et al., 2015]

Remarks

- These architectures easily contain a number of parameters of the order of 10^7 (28 million for U-Net)
- Their optimization might be difficult
- Depending on the complexity of the segmentation task, the number of filters or the number of layers can be reduced

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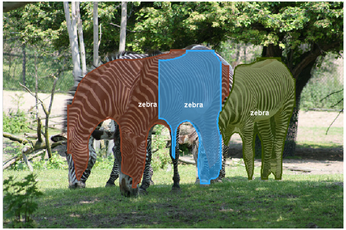
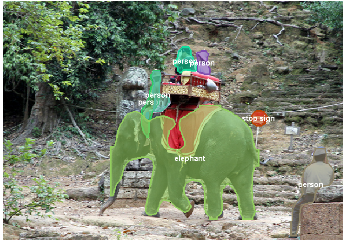
COCO: common objects in context [Lin et al., 2014]

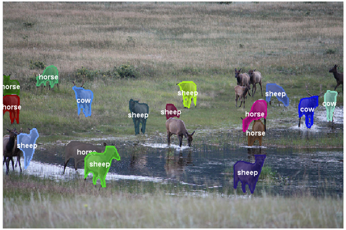
- 2 million objects, from 80 categories, in $300\,000$ images

Winner 2016: Fully Convolutional Instance-aware Semantic Segmentation (Microsoft) [Li et al., 2016]

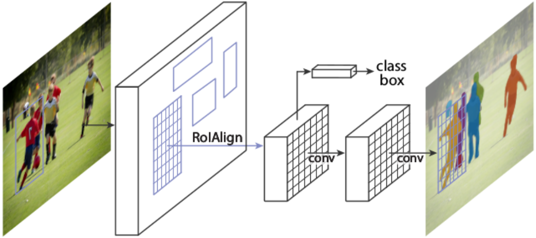
COCO instance segmentation challenge: examples of 2016 winner results



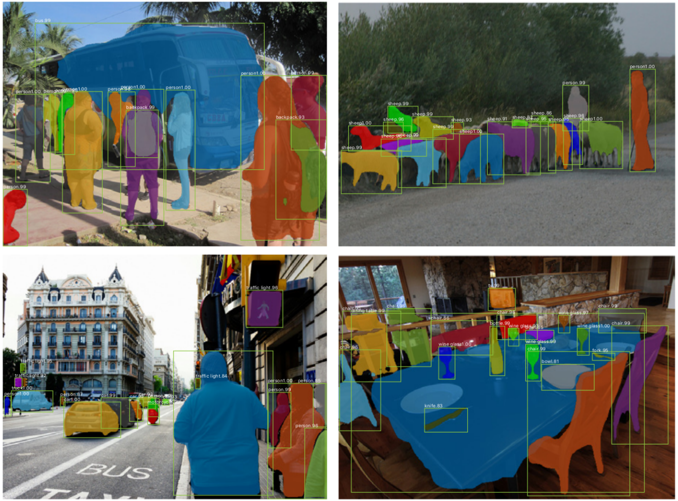




State of the art on the COCO database: Mask R-CNN [He et al., 2017]

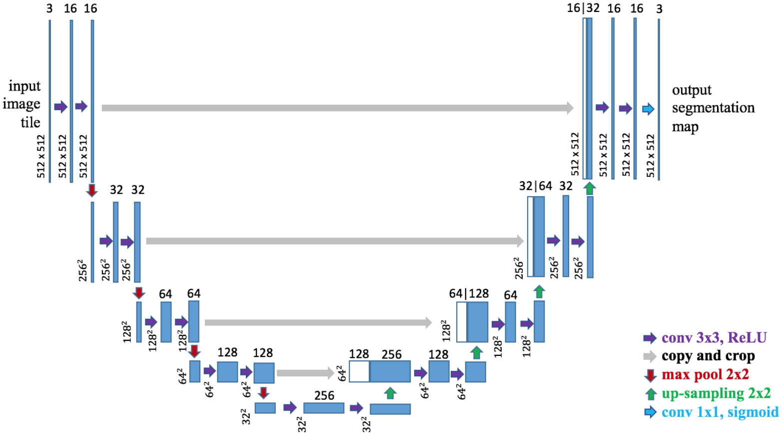


Mask R-CNN on the COCO database

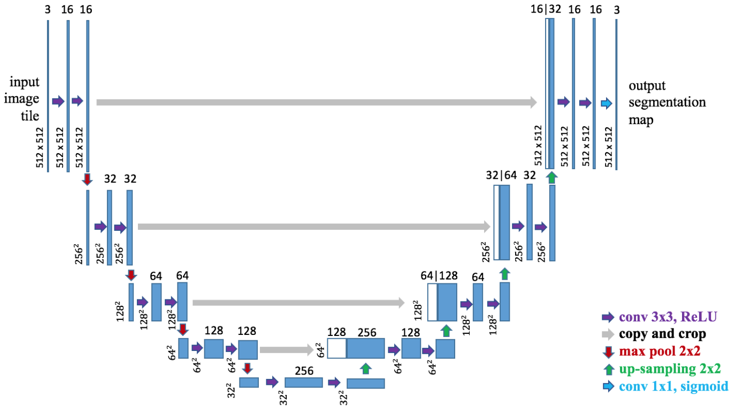


Contents

U-Net architecture [Ronneberger et al., 2015]



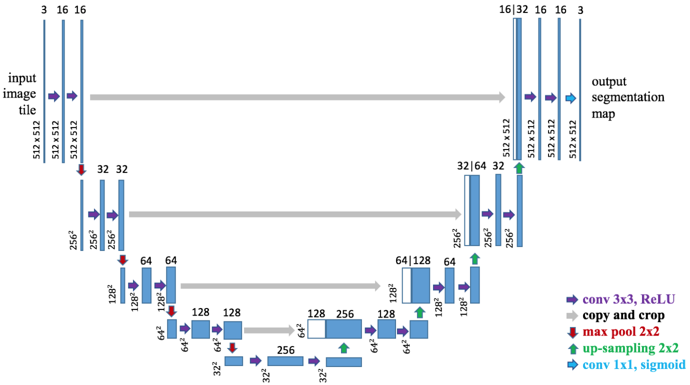
U-Net architecture [Ronneberger et al., 2015]



Quizz

- Size and number of channels of input images?
- Segmentation into how many regions?

U-Net main ideas



- Encoding branch inspired by classification nets
- Decoding branch is symmetrical
- Skip connections

U-Net improvements

- Convolutions with stride 2 instead of max-pooling in the encoder
- Transposed convolutions instead of simple up-sampling in the decoder
- Batch normalization
- Using an already optimized classification network as backbone for the encoder

Contents

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- Solution: extract fixed-sized crops from your training set:
  - make them as large as possible, to reduce border effects

Loss functions for image segmentation

- $\hat{\mathbf{y}} = (\hat{\mathbf{y}}_i)$: network output
- $\mathbf{y} = (\mathbf{y}_i)$: binary expected output
- We suppose that all $\hat{\mathbf{y}}_i$ are in $[0, 1]$
- We want the $\hat{\mathbf{y}}$ to be *as close as possible* to $\mathbf{y}$

Loss functions for image segmentation

A loss function inherited from image classification

- Cross-entropy: $-\sum_i y_i \log(\hat{y}_i)$

Measures used in image processing

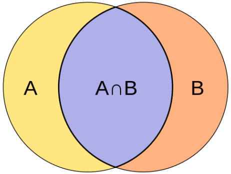
Let A and B be two sets, not simultaneously empty.

Dice coefficient

$$D(A,B) = \frac{2|A \cap B|}{|A| + |B|}$$

Jaccard index

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$



Properties

- $\forall A, B : 0 \leq J(A,B) \leq D(A,B) \leq 1$
- If $A = B$, then $D(A,B) = J(A,B) = 1$
- If $A \cap B = \emptyset$, then $D(A,B) = J(A,B) = 0$

Credits: Public domain (Wikipedia.org)

Generalization to $[0, 1]$

Jaccard index

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

But $\mathbf{y}$ and $\hat{\mathbf{y}}$ are in $[0, 1]^n \ldots$

Generalization to $[0, 1]$

Jaccard index

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

But $\mathbf{y}$ and $\hat{\mathbf{y}}$ are in $[0, 1]^n \dots$

Jaccard similarity

$$J(\mathbf{y}, \hat{\mathbf{y}}) = \frac{\sum_i y_i \hat{y}_i}{\sum_i y_i + \sum_i \hat{y}_i - \sum_i y_i \hat{y}_i}$$

($\mathbf{y}$ and $\hat{\mathbf{y}}$ are not simultaneously equal to 0)

Corresponding loss function

Jaccard loss

$$j(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \frac{\sum_i y_i \hat{y}_i}{\sum_i y_i + \sum_i \hat{y}_i - \sum_i y_i \hat{y}_i + \epsilon}$$

Constant ϵ, which is typically “small”, keeps the denominator “far enough” from zero.

Corresponding loss function - variant

Jaccard loss

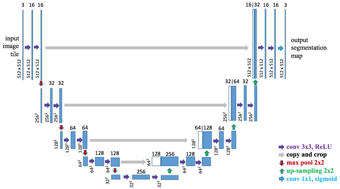
$$j_2(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \frac{\sum_i y_i \hat{y}_i}{\sum_i y_i^2 + \sum_i \hat{y}_i^2 - \sum_i y_i \hat{y}_i + \epsilon}$$

This version works slightly better than the first [Duque-Arias et al., 2021].

Conclusion on loss functions

- Use the Jaccard loss as base line for segmentation problems.
- Note that these losses compute their values pixel-wise: they do not take into account any structure (for example, continuity).
- Working on specific losses enforcing structure is an interesting research path.

Applying fully-convolutional networks



Case study

Suppose that we have satisfactorily optimized this U-Net model, using images of size 512×512 during training. Now I want to apply this model to a new image, of size 1000×1000. How should I proceed?

1. Resize the image?
2. Cut it into 512×512 crops, predict and stitch the results together?
3. Pad the image.

Segment anything

https://ai.facebook.com/blog/segment-anything-foundation-model-image-segmentation/

Image segmentation: a solved problem?

- Progress in image segmentation since 2012 has been enormous
- Several complex problems have now satisfactory solutions
- Training can be a problem (large annotated databases, difficult optimization)

Some research subjects

- Optimization - a very general, and essential, subject
- Making training databases as small as possible
- Specific losses
- Taking *a priori* structural information into account

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